

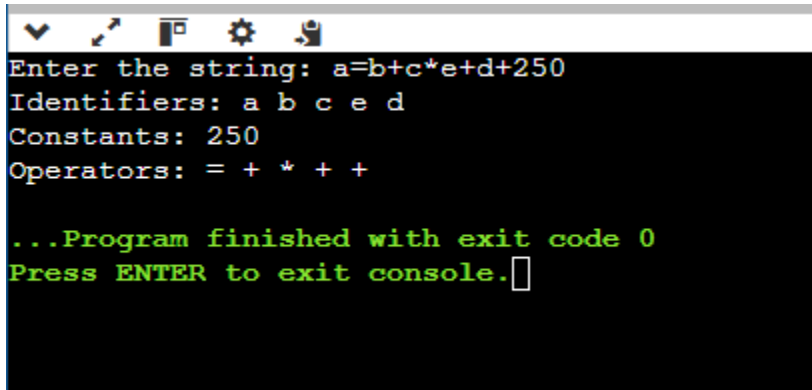
Experiment - 1

Develop a lexical Analyzer to identify identifiers, constants, operators using C program.

CODE :

```
#include <stdio.h>
#include <ctype.h>
int main()
{
    char b[100];
    int i;
    printf("Enter the string: ");
    scanf("%[^\n]", b);
    printf("Identifiers: ");
    for (i = 0; b[i]; i++)
        if (isalpha(b[i]))
            printf("%c ", b[i]);
    printf("\nConstants: ");
    for (i = 0; b[i]; i++)
        if (isdigit(b[i])) {
            while (isdigit(b[i])) {
                printf("%c", b[i++]);
            }
            printf(" ");
            i--;
        }
    printf("\nOperators: ");
    for (i = 0; b[i]; i++)
        if (b[i] == '+' || b[i] == '-' ||
            b[i] == '*' || b[i] == '=')
            printf("%c ", b[i]);
    return 0;
}
```

OUTPUT :



```
Enter the string: a=b+c*e+d+250
Identifiers: a b c e d
Constants: 250
Operators: = + * + +

...Program finished with exit code 0
Press ENTER to exit console.
```

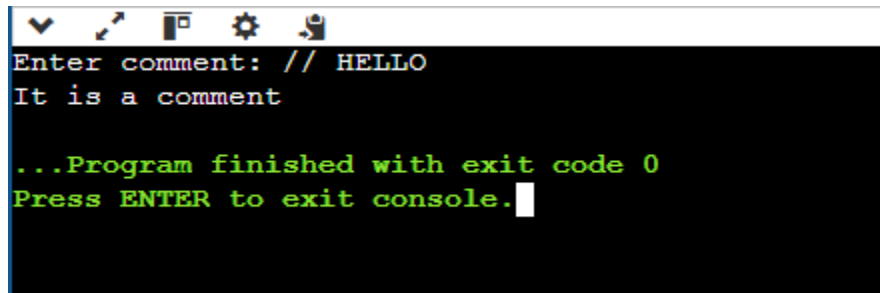
Experiment - 2

Develop a lexical Analyzer to identify whether a given line is a comment or not using C

CODE :

```
#include <stdio.h>
#include <string.h>
int main()
{
    char com[100];
    printf("Enter comment: ");
    fgets(com, sizeof(com), stdin);
    if (com[0] == '/' && com[1] == '/')
        printf("It is a comment");
    else if (com[0] == '/' && com[1] == '*'
        && strstr(com, "*/"))
        printf("It is a comment");
    else
        printf("It is not a comment");
    return 0;
}
```

OUTPUT :

A screenshot of a console window with a black background and white text. The window has a title bar with standard icons (minimize, maximize, close, settings, and a user icon). The text inside the console reads: "Enter comment: // HELLO", "It is a comment", "...Program finished with exit code 0", and "Press ENTER to exit console." followed by a white cursor block.

```
Enter comment: // HELLO
It is a comment

...Program finished with exit code 0
Press ENTER to exit console.
```